MINI PROJECT DOCUMENTATION

AI Fake Document Detection System

DocVerify AI

An AI-assisted web application for document authenticity risk screening

Project Type	B.Tech AI & Machine Learning — Mini Project
Development Platform	Base44 AI
Application Type	Web Application
Domain	Artificial Intelligence / Document Security
Developed By	Jakkula Anusha

Live Demo

https://hypersonic-doc-verify-pulse.base44.app

Prepared for academic mini-project documentation and GitHub/portfolio presentation.

1. Abstract

Digital documents such as academic certificates, invoices, offer letters, receipts, and other official records can be modified using modern editing tools. Manual inspection of such documents can be time-consuming and may miss subtle inconsistencies. **DocVerify AI** is a web-based AI-assisted document screening system designed to identify potential signs of document manipulation and inconsistency.

The application allows users to upload supported documents and receive an authenticity **risk assessment** based on available textual, visual, structural, and metadata-related indicators. The system presents a risk score, risk level, suspicious indicators, explanations, and recommended actions in an easy-to-understand dashboard.

The project was developed using **Base44 AI** as the application-building platform. It demonstrates how AI concepts can be integrated into a practical web application for document security. The system is intended as a screening and decision-support prototype and does not provide legal proof that a document is authentic or fraudulent.

2. Problem Statement

Digital document manipulation has become increasingly accessible. Educational institutions, recruiters, businesses, and individuals may need to perform an initial verification of documents before accepting them. Traditional manual verification requires time, attention, and sometimes specialized expertise.

The proposed system addresses this challenge by providing an AI-assisted first-level screening mechanism that can identify potentially suspicious characteristics and summarize them as an understandable risk assessment.

3. Objectives

The major objectives of the project are:

1. Develop a responsive web application for AI-assisted document screening.

2. Allow users to upload documents through a simple interface.

3. Analyze available textual, visual, structural, and metadata-related characteristics.

4. Generate an authenticity risk score and risk category.

5. Identify and explain potentially suspicious indicators.

6. Maintain analysis history for authenticated users.

7. Provide a structured analysis/report view suitable for academic demonstration.

8. Create an extensible foundation for advanced OCR, computer vision, and deep-learning models.

4. Existing System and Proposed System

Existing approach: Document verification is often performed manually or through isolated verification services. Manual inspection can be slow and inconsistent, particularly when large numbers of documents must be screened.

Proposed approach: DocVerify AI provides a single web interface where users can upload a document, initiate analysis, view risk indicators, understand the reasons for the assessment, and access a structured report.

Aspect	Traditional / Manual	DocVerify AI
Verification	Manual inspection	AI-assisted screening
Speed	Time-consuming	Fast initial assessment
Output	Human judgement	Risk score + indicators
Explanation	Depends on reviewer	Structured AI explanation
History	May be manual	Digital analysis history

5. Proposed Methodology

The application follows a multi-stage workflow. The uploaded document is processed and the available signals are organized into analysis categories. These signals are then combined into an overall risk assessment and presented to the user.

1	Upload	User uploads a PDF or supported image document.
2	Process	Document content and available properties are prepared for analysis.
3	Analyze	Text, visual, structural, and metadata indicators are examined.
4	Assess	Potential anomalies are converted into a risk assessment.
5	Explain	The system presents indicators, severity, and explanations.
6	Report	The user can review and retain the analysis result.

6. System Modules

Landing Page

Introduces DocVerify AI, its purpose, features, workflow, and supported document types.

Authentication

Provides login and registration functionality for user-specific access.

Dashboard

Displays analysis statistics, risk distribution, and recent activity.

Document Upload

Provides drag-and-drop/browse functionality and document-type selection.

Document Analysis

Organizes analysis around text, visual, metadata, and structural indicators.

Risk Assessment

Generates a 0–100 risk score and maps it to a risk category.

Results & Explanation

Shows suspicious indicators, severity, confidence, explanations, and recommendations.

Analysis History

Allows users to search, filter, and review previous analyses.

Report

Presents the analysis in a structured report format.

7. Technologies and Tools

Category	Technology / Concept
Application Builder	Base44 AI
Frontend	Responsive web UI generated and configured through Base44
AI Concepts	Document analysis, anomaly detection, NLP/computer-vision concepts
Data	User, document, analysis, and report records
Visualization	Dashboard cards, risk indicators, charts, and progress states
Deployment	Base44-hosted web application

8. AI/ML Concepts

Anomaly Detection: Detects patterns that differ from expected document characteristics.

Natural Language Processing: Supports analysis of extracted text and textual consistency.

Computer Vision: Provides a conceptual basis for examining document images, layouts, and visual inconsistencies.

Feature Analysis: Considers multiple document characteristics rather than relying on a single signal.

Risk Classification: Converts analysis signals into interpretable risk categories.

Score	Risk Level
0–30	Low
31–60	Medium
61–80	High

Score	Risk Level
81–100	Very High

9. Key Features

1. Document upload with file type and size validation.

2. AI-assisted analysis workflow with progress indicators.

3. Risk score from 0 to 100.

4. Low, Medium, High, and Very High risk categories.

5. Suspicious indicator cards with severity.

6. Document analysis history.

7. Dashboard statistics and risk distribution.

8. Structured report presentation.

9. Responsive design for desktop, tablet, and mobile.

10. Privacy-oriented user-specific access.

10. Sample Analysis Result

The following is an illustrative example of how the application can present a screening result:

Document	Academic_Certificate.pdf
Risk Score	68 / 100
Risk Level	High
Indicator 1	Font inconsistency — High severity
Indicator 2	Image-quality difference — Medium severity
Indicator 3	Layout inconsistency — Medium severity
Metadata	Unavailable / not sufficient for assessment
Recommendation	Manual verification with the issuing organization

11. Advantages

• Provides a fast first-level screening workflow.

• Combines multiple categories of document indicators.

• Presents technical analysis in a simple visual format.

• Useful as an academic demonstration of applied AI.

• Can be extended to more document types and advanced models.

• Provides a reusable web-based interface rather than a command-line prototype.

12. Limitations

• AI-based screening cannot guarantee that a document is authentic or fraudulent.

• Low-quality scans and images can reduce the reliability of analysis.

• Some files may not contain useful metadata.

• Advanced manipulation techniques may not be detected.

• Official verification by the issuing organization may still be necessary.

• Actual detection accuracy depends on the underlying analysis/model implementation and available data.

13. Future Enhancements

• Advanced OCR for scanned and multilingual documents.

• Deep-learning models for image-forensics and tampering detection.

• Pixel-level suspicious-region localization.

• Explainable AI for clearer feature-level reasoning.

• Digital signature and QR-code verification.

• Blockchain-based certificate verification.

• Integration with authorized issuer databases.

• Support for multiple Indian languages.

• Benchmarking against labeled real and manipulated document datasets.

14. Conclusion

DocVerify AI demonstrates a practical application of artificial intelligence concepts to the problem of digital document screening. The project provides an end-to-end web workflow in which users upload a document, the application organizes available analysis signals, and the result is presented through a risk score, indicators, explanations, and recommendations.

The project also demonstrates how **Base44 AI** can be used as an application development platform to rapidly create and deploy a functional AI-oriented web prototype. With further development, the system can incorporate advanced OCR, computer vision, deep learning, digital signatures, and trusted issuer verification.

15. Project Details

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Live Application	https://hypersonic-doc-verify-pulse.base44.app

16. Disclaimer

DocVerify AI is an academic/project prototype for AI-assisted document screening. Its risk scores and indicators should not be treated as definitive proof that a document is fake, fraudulent, authentic, or legally valid. For official verification, users should contact the relevant issuing institution or authorized authority.

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